

Substations		Technical Procedure	
Equipment Power Transformers		Document No:	T-027-r0
		Issued to:	Networks
		Status:	APPROVED
Procedure: Turns Ratio Test		Approved Date:	05 Feb 2001
		Date to be Reviewed:	Feb 2006

Introduction

This document lists the steps for testing the turns ratio for the power transformers.

Safety Precautions

- Isolate the transformer out of service earthen
- A word order permit must be issued.
- The qualified test staff should carry the suitable safety category
- Safety fence (rope) with caution marks surrounding the work area,
- Disassembly all clamps from bushings (High Voltage side, Low Voltage side & tertiary if exists)
- Connect the earth of tester to the common earth of substation.
- Don't misapply the High leads and Low leads of the test equipment to the transformer bushings.
This will cause damage to the test equipment.
- Keep test leads apart from the buahings.

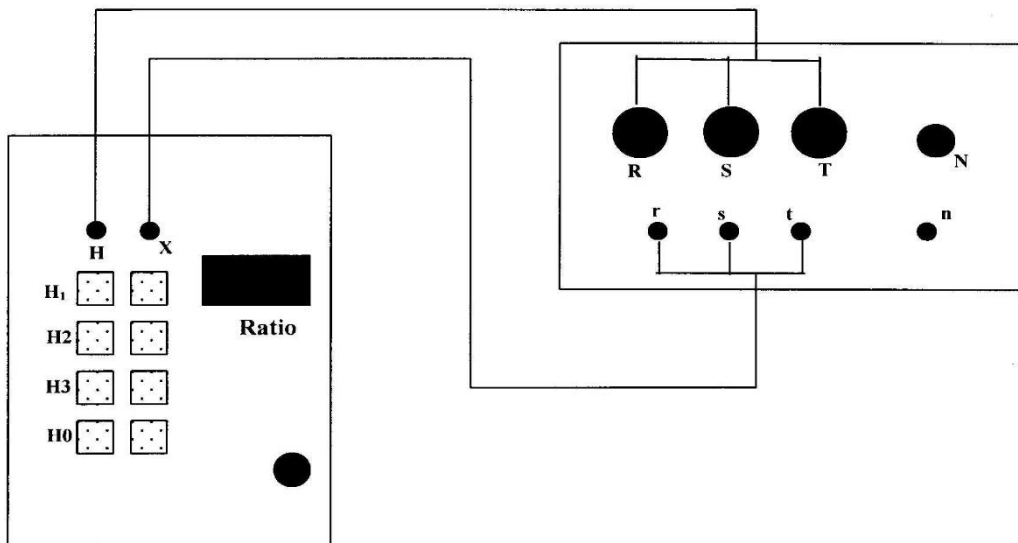
Tools and Equipment

- Test equipment type DRX 2000, The company address: Electrical Testing Instruments Scarbough, Ontario Canada M1V1E7 telephone 416-292-8181, fax: 416-292-8188

Work to be Carried Out

1. Earth the test equipment with common earth of the substation.

2. Connect the HV three-phase test leads (H1, H2, and H3) to the three phases HV bushings (R, S, T) of the transformer. Notice that H₀ is neutral point with the neutral bushing.
3. Connect the LV three phase test leads (X1,X2,X3) to the three phase LV bushing of the transformer notice that X₀ is neutral point with the neutral bushing.
4. Run the test between each as the following steps:
 - Switch on the tester and press H₁-H₀
 - Press X₁-X₀
 - Press the test button Record the result.
5. Repeat the previous test procedure with rest phases H2, H0..... Press X₂-X₀
6. Repeat the previous test procedure with rest phases H3, H1 Press X₃-X₀
7. Follow the test procedures as the next drawing
8. Record the result In the Attached Test sheet



3B-Turns Ratio Test For Two winding transformer

TAP	(H ₁ -H ₂) An-an (X ₁ -X ₀)	(H ₂ -H ₃) Bn-bn (Y ₂ -X ₀)	(H ₃ -H ₁) Cn-cn (X ₃ -X ₀)
1			
2			
3			
4			
5			

6			
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10			
11			
12			
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21			

Ratio Test Results (for Two Winding Transformer)

Tap Position	Calculated	Phase A	Phase B	Phase C
		$(H_1-H_2)/(X_1-X_0)$	$(H_2-H_3)/(X_2-X_0)$	$(H_3-H_1)/(X_3-X_0)$
1				
2				
3				
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21				

Ratio Test Results (for Three Winding Transformer)

Tap Position	Calculated	Phase A	Phase B	Phase C
		$(H_1-H_2)/(X_1-X_0)$	$(H_2-H_3)/(X_2-X_0)$	$(H_3-H_1)/(X_3-X_0)$
1				
2				
3				
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Location:
Checked by:

Equipment code:
Date:

Signatur